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Blame and forgiveness judgements among children, adolescents and adults with autism

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ABSTRACT We compared the capacity of children, adolescents and adults with and without autism to use (a) intent and severity of consequences information for attributing blame to an offender, and (b) intent and apologies information for inferring willingness to forgive. Participants were presented with two sets of six scenarios obtained by combination of intent and severity (or apology) information, and instructed to indicate appropriate levels of blame (or willingness to forgive). In the blame condition, persons with autism were able to consistently use intent information but not to the same degree as their comparison counterparts. In the forgiveness condition, intent was not taken into account for judging by persons with autism, irrespective of their age.

KEYWORDS *autism, blame, forgiveness*

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Blaming and avoiding blame, forgiving and seeking forgiveness are common phenomena implied in every level of social interaction (Anderson, 2008, in press). Blaming a person for his or her harmful actions naturally depends on the damage that has been caused, but also depends on that person's intent. In the same way, forgiving a person for their harmful actions depends on several factors, among them that person's initial intent and their subsequent behaviour (Anderson, in press). As (a) persons with autism have been shown to have difficulty understanding and interpreting others' intents (e.g. D'Entremont and Yazbek, 2007), which has been associated with reduced 'theory of mind' abilities (Young et al., 2007), and (b) intent is a central and omnipresent determinant of moral judgements (Cushman, 2008), examining the way in which persons with autism use intent information in combination with other information (e.g. severity of consequences, presence of apologies) for making moral judgements is, as suggested by Young et al. (2007), an important issue.

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Moral judgement in persons with autism has been investigated by several authors. Blair (1996) examined the ability of children with autism to distinguish moral transgressions (e.g. hitting others) from conventional transgressions (e.g. talking in class). In the first session, 20 children with autism were presented with several stories about transgressions at school, and they were asked whether the transgression was permissible or not. In the second session, these children were told that, in this school, a teacher has declared that everything was permitted. The children were presented with the same stories again and they were asked the same question. The children made a distinction between the two types of transgression. The permissibility scores of the conventional transgressions significantly dropped from the first to the second condition whereas the permissibility score of the moral transgressions remained similar. Blair (1996) concluded that at least relatively high-functioning children with autism show sensitivity to the distress of others.

Grant et al. (2005) examined the ability of children with autism for making moral judgements. They presented their participants with pairs of stories about transgressions. These stories were constructed as a function of two factors: intent of the harmful act (accidental versus deliberate), and severity of the harmful act (damage to property versus injury to a person). The participants were instructed to decide which transgression in each pair was naughtier and to verbally justify their decision. When presented with pairs of scenarios that had similar outcomes (e.g. personal injury) but different motives (ill motive versus neutral motive), children with autism were able to judge culpability on the basis of intent information. When presented with scenarios that had different outcomes but similar motives, children with autism were able to judge injury to persons as more severe than damage to property. Justifications offered were, however, of poor quality, and were often mere reiterations of the stories. The present study differs from Grant et al.'s (2005) study in several aspects. Firstly, three groups of persons with autism were considered – children, adolescents and adults – as well as three groups of comparison participants matched with respect to chronological age and three groups of comparison participants matched with respect to developmental age. As suggested by Steele et al. (2003), developmental changes in theory of mind abilities may occur among persons with autism. As a result, by considering three very different age groups of persons with autism and comparison participants, we were in a position to examine whether development and autism interacted with moral judgement.

Secondly, as children with autism may have difficulty with verbalization (see Grant et al., 2005), the procedure we used did not require verbal responses. Stories about transgressions were constructed as a function of the same two factors as in the study by Grant et al. (2005) (intent of the

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harmful act and severity of its consequences) but a complete orthogonal design was used, and judgements about the level of blame attributed to the offender were simply recorded on a linear scale ranging from no blame at all to strong level of blame. No comparison between transgression situations was made, and verbalizations were kept to a minimum.

Thirdly, we considered not just one but two moral situations, one involving blame and one involving forgiveness. Forgiving is an apparently more complex issue than blaming. In the case of blaming, the more severe the consequences and the clearer the intent, the stronger the blame; that is, the two relationships (consequences-blame and intent-blame) are direct ones. In the case of forgiving, the more complete the apologies, the stronger is the willingness to forgive, but the clearer the intent, the weaker is the willingness to forgive; that is, one of the two relationships (intent-forgiveness) is an inverse one (Mullet et al., 2007). Using inverse relationships for making judgements has been shown to be more difficult than using direct relationships, especially in children (Lafon et al., 2004). As a result, children with autism's ability to use intent information may vary depending on the moral situation even if not much difference is evidenced among comparison participants.

The first hypothesis was that children, adolescents and adults with autism should differ from their comparison counterparts in the way that they use the intent information in both sets of stories (blame and forgiveness). This hypothesis was based on a host of findings showing that persons with autism have trouble interpreting another person's intent (e.g. D'Entremont and Yazbek, 2007). It was also based on the findings of Grant et al. (2005): when presented with pairs of stories that varied in both outcomes and motives, children with autism did not perform as well as their comparison counterparts. It was expected that the effect of intent on blame judgement and on forgiveness judgement should be significantly weaker among persons with autism than among persons without autism. The second hypothesis was that a developmental change should manifest itself among participants with autism that should parallel the developmental change observed among their comparison counterparts. Namely, in both situations (blame and forgiveness), the effect of intent should be significantly stronger among adults with autism than among children with autism.

Method

Participants

The participants were 25 persons with autism (10 children aged 7 to 14 years, 9 adolescents aged 15 to 18 years, and 6 adults aged 22 to 36 years),

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25 persons of the same mental age without autism, and 25 persons of the same chronological age without autism (20 males and 5 females in each group). All persons with autism received a diagnosis of autism by a qualified psychiatrist. The scales used were the French adaptation (Rogé et al., 2010) of the Autism Diagnostic Interview (Rutter et al., 2003), and the French adaptation (Rogé et al., 2009) of the Autism Diagnostic Observation Schedule (Lord et al., 2001).

Material

The material was composed of two series of six scenarios. In the blame scenarios, a transgression was committed, which was intentional or not (accidental or deliberate) and had more or less serious consequences (not severe, medium and severe). The orthogonal combination of the two factors resulted in six scenarios. An example of blame scenario (no intent, severe consequences) is: 'Paul uses John's portable radio. Accidentally, Paul lets the radio drop to the ground. Because of the shock, the radio is broken. It cannot be repaired. In your opinion, what level of blame does Paul deserve?'. Another example of blame scenario (clear intent, no consequences) is: 'Peter is jealous of Jim's portable radio. He asks Jim let him use the receptor. Intentionally, Peter lets the radio fall down on the ground. Despite the shock, the radio is not broken. It keeps functioning. In your opinion, what level of blame does Peter deserve?'. Under each scenario a response scale was printed, ranging from 'No blame at all' to 'Strong blame'. (The complete set of scenarios is available from the authors.)

In the forgiveness scenarios, the transgressions were intentional or not, and were followed by more or less clear apologies (no apologies, indirect apologies and direct apologies). An example of forgiveness scenario (intent to harm, direct apologies) is: 'Patrick and Tom play with a ball. Patrick runs after John to get the ball. Patrick trips Tom up on purpose. Tom falls down and hurts his knees very badly. Immediately, Patrick regrets what he has done, and apologises several times to Tom. If you were Tom, to what extent do you feel you would be willing to forgive to Patrick?'. Another example of forgiveness scenario (no intent to harm, indirect apologies) is: 'Sam and Henry play with a ball. Sam runs after Henry to get the ball. Sam trips Henry up by accident. Henry falls down and hurts his knees very badly. The day after, Patrick asks Oliver, who is a friend of both of them, to tell Henry that he regrets what has happened, and apologises. If you were Henry, to what extent do you feel you would be willing to forgive to Sam?'. The response scale ranged from 'Sure I wouldn't forgive' to 'Sure I would forgive'.

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Procedure

Firstly, as recommended by Anderson (1982), the experimenter explained to each participant what was expected from him or her, in the so-called familiarization phase. The participant read out loud the six stories in which a person committed a harmful act against another person. After each story was read, the experimenter reminded them of the critical items of information. The participants then rated the degree of blame they would attribute to the offender or of their personal willingness to forgive if they were the victim. They were allowed to compare their responses and change them.

In a second, experimental, phase, the same six stories were presented again (in a different order for each participant), and the participants made their ratings. It was no longer possible to compare responses or to go back and make changes. Children were tested individually and there was no time limit. For half of the participants the blame scenarios were presented first, and for the remaining half, the forgiveness scenarios were presented first.

Results

For each series of six scenarios in the second, experimental, phase, the distance was measured between the left anchor (no blame or no forgiveness) and each answer. All subsequent analyses were based on these measures of distance.

Blame stories

The data were analysed, at the group level, by performing analyses of variance (ANOVAs), and by constructing detailed graphs. Two separate ANOVAs were conducted, one contrasting persons with autism and comparison participants of the same developmental age, and the other contrasting persons with autism and comparison participants of the same chronological age. Two separate ANOVAs were conducted because the two comparison groups were too different to be meaningfully combined in the same ANOVA, and statistical power was strong.

The design of the first ANOVA was participant's group (persons with autism versus comparison participants of the same developmental age) × age (children, adolescents, adults) × intent × severity, 2 × 3 × 2 × 3. Table 1 (left side) shows the detailed results. Figure 1 (all nine panels) shows the effect of each factor (within and between subjects). An index of curve separation was computed by subtracting the lowest mean value from the highest mean value. Curve separation between intent and no intent was 7.29: when there was clear intent the level of blame was, for most groups, higher than when the act had been accidental. Curve separation between severe consequences and no consequences was 7.43: when the consequences were severe the level of blame was higher than when they were not severe.

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Table 1 **Results of the analyses of variance conducted with the blame scenarios**

		Persons with autism versus comparison group (developmental)				*Persons with autism versus comparison group (chronological)*			
Factors	*d.f.*	*MS*	F	p	Eta^2	*MS*	F	p	Eta^2
Group (G)	1	1.24	0.03	NS	0.00	2.12	0.05	NS	0.00
Age (A)	2	345.77	7.90	.01	0.26	385.38	9.67	.01	0.31
Intent (I)	1	3 400.84	72.21	.01	0.62	4 431.95	83.04	.01	0.65
Consequences (C)	2	1 248.45	44.12	.01	0.50	1 260.51	46.07	.01	0.51
G × A	2	38.58	0.88	NS	0.04	99.15	2.49	NS	0.10
G × I	1	252.96	5.37	.05	0.11	472.42	8.85	.01	0.17
A × I	2	284.56	6.04	.01	0.22	513.98	9.63	.01	0.30
G × C	2	21.30	0.75	NS	0.02	45.60	1.67	NS	0.04
A × C	4	5.74	0.20	NS	0.01	2.21	0.08	NS	0.00
I × C	2	10.31	1.03	NS	0.02	2.67	0.34	NS	0.01
G × A × I	2	5.49	0.12	NS	0.01	31.86	0.60	NS	0.03
G × A × C	4	30.08	1.06	NS	0.05	27.24	1.00	NS	0.04
G × I × C	2	7.40	0.74	NS	0.02	1.46	0.19	NS	0.00
A × I × C	4	21.79	2.18	NS	0.09	12.84	1.65	NS	0.07
G × A × I × C	4	11.11	1.11	NS	0.05	3.85	0.49	NS	0.02

NS: not significant.

The separation of the curves increased from the top row (children) to the bottom row (adults). In other words, the effect of intent was stronger among adults (10.09) than among children (3.59). Finally, the curves were less separated among the persons with autism (left panels, 5.30) than they were among the comparison participants (centre panels) (9.28). Interestingly, both the age × consequences interaction and the age × group × intent interaction were not significant. A planned comparison conducted on the data on the group of persons with autism showed that the linear × linear component of the age × intent interaction was marginally significant ($p = .07$). The pattern of effects observed in the second ANOVA (see Table 1, right side) where comparison participants of the same chronological age were substituted for comparison participants of the same developmental age was extremely similar to the one observed in the first ANOVA.

Forgiveness stories

ANOVAs with similar designs to those of the blame stories were conducted (Table 2). The design of the first ANOVA was participant's group (persons with autism versus comparison participants of the same developmental age) × age (children, adolescents, adults) × intent × apologies, 2 × 3 × 2 × 3. Intent and apologies had significant effects. As shown in Figure 2, curve separation between intent and no intent for the forgiveness stories was 3.59: for most groups, when there was clear intent, willingness to forgive was lower than when the act was accidental. Curve separation

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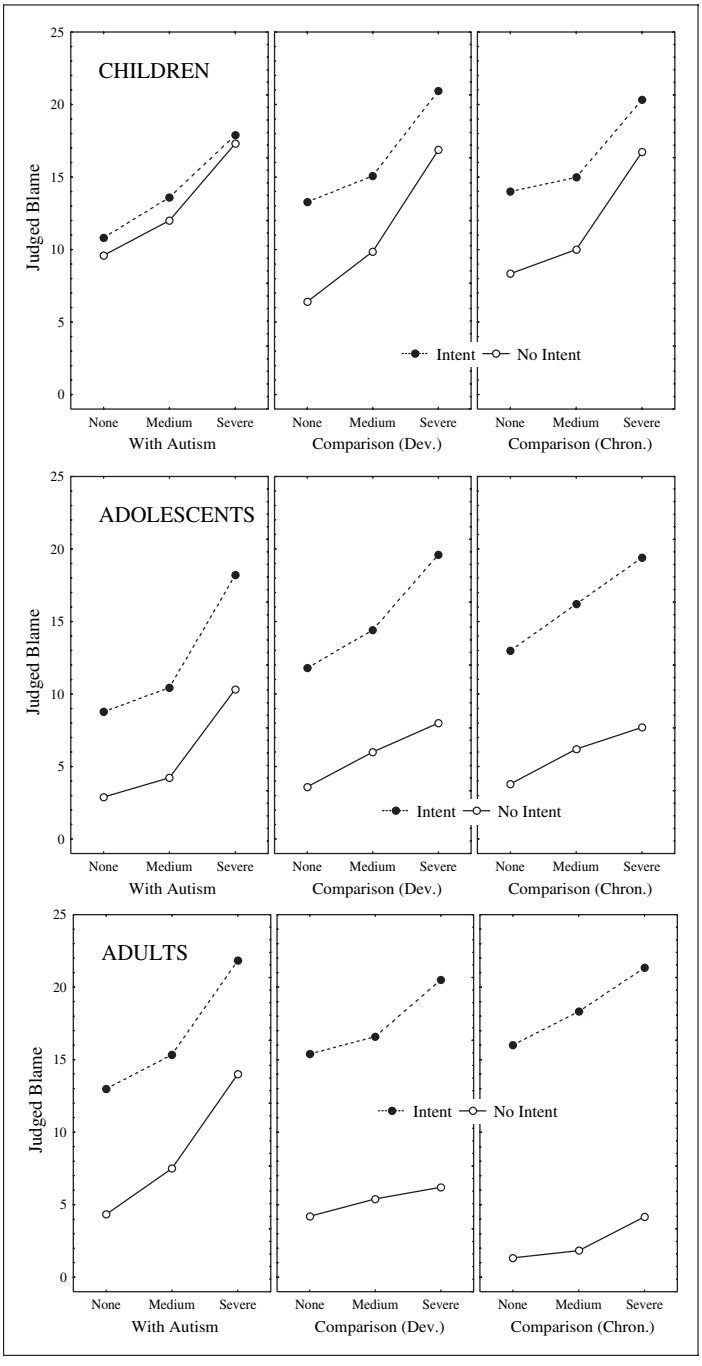

Figure 1 **Blame judgements as a function of intent and severity information. Dev.: developmental, Chron.: chronological**

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Table 2 **Results of the analyses of variance conducted with the forgiveness scenarios**

		Persons with autism versus comparison group (developmental)				*Persons with autism versus comparison group (chronological)*			
Factors	*d.f.*	*MS*	F	p	Eta^2	*MS*	F	p	Eta^2
Group (G)	1	41.88	0.84	NS	0.02	23.00	0.50	NS	0.00
Age (A)	2	146.70	2.95	NS	0.12	259.99	5.70	.01	0.21
Intent (I)	1	826.77	18.78	.01	0.30	723.69	18.81	.01	0.30
Apologies (P)	2	2799.94	87.46	.01	0.67	3361.59	120.52	.01	0.73
G × A	2	115.86	2.33	NS	0.10	116.87	2.56	NS	0.10
G × I	1	371.95	8.45	.01	0.16	285.35	7.41	.01	0.14
A × I	2	61.40	1.39	NS	0.06	30.78	0.80	NS	0.04
G × P	2	51.69	1.61	NS	0.04	91.66	3.29	.05	0.07
A × P	4	17.65	0.55	NS	0.02	58.31	2.09	NS	0.09
I × P	2	1.04	0.07	NS	0.00	2.18	0.14	NS	0.00
G × A × I	2	7.58	0.17	NS	0.01	22.69	0.59	NS	0.03
G × A × P	4	87.34	2.73	NS	0.11	50.40	1.81	NS	0.08
G × I × P	2	7.77	0.53	NS	0.01	11.28	0.74	NS	0.02
A × I × P	4	36.40	2.49	NS	0.10	35.05	2.28	NS	0.09
G × A × I × P	4	22.95	1.57	NS	0.07	26.48	1.73	NS	0.07

NS: not significant.

between direct apologies and no apologies was 11.46: when direct apologies were present, willingness to forgive was higher than when they were absent.

From the top row to the bottom row, the separation of the curves generally increased. The effect of intent was apparently stronger among adults (4.12) than it was among children (1.89), but the age × intent interaction was not significant. Finally, among the persons with autism (left panels), the curves were less separated (1.18) than they were among the comparison participants (centre panels) (6.01). Interestingly, both the age × apologies interaction and the age × group × apologies interaction were not significant. As for the second ANOVA (2 × 3 × 2 × 3) where comparison participants of the same chronological age were substituted by comparison participants of the same developmental age, the pattern of effects was extremely similar.

Discussion

In the blame condition, comparison participants used, as expected, intent and severity information for inferring level of blame. This finding is consistent with previous findings in the literature (Anderson, 2008, in press; Cushman, 2008; Hommers and Anderson, 1991; Surber, 1977, 1982). Persons with autism also used intent and severity information but intent was given significantly less weight than it was given among comparison participants. This result is consistent with Grant et al.'s (2005) findings that

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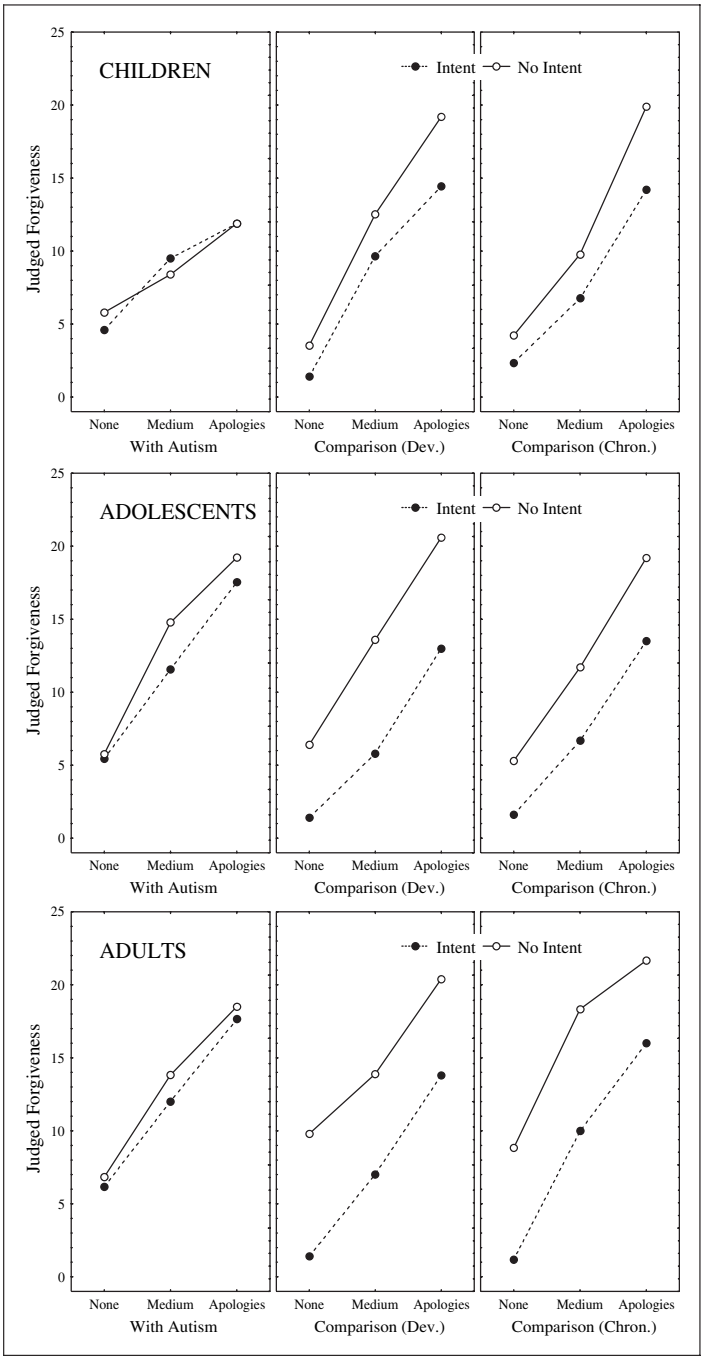

Figure 2 **Willingness to forgive judgements as a function of intent and apologies information. Dev.: developmental, Chron.: chronological**

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children with autism were able to use intent information, and with Leslie and Mallon's (2006) findings that basic moral judgement was substantially intact in children with autism. Our findings are, nevertheless, also consistent with previous findings showing that participants with autism have trouble at interpreting others' intents (e.g., D'Entremont and Yazbek, 2007), and with Grant et al.'s (2005) findings that when presented with pairs of scenarios with different outcomes and with different motives, children with autism did not perform as well as their comparison counterparts. In sum, despite being able to consistently use intent information, persons with autism were not able to do this to the same degree as their comparison counterparts. There was, however, no detectable difference in the use of severity information, which is consistent with Blair's (1996) findings showing that children with autism are able to distinguish severe moral transgressions from minor conventional transgressions.

Intent was, as expected, given more weight among comparison adults than among comparison children, and is consistent with previous findings (Anderson, 2008; Leon, 1982; Surber, 1982). This developmental trend was also found among persons with autism, and it seemed not to differ much between comparison participants and persons with autism (the age × group × intent interaction was far from being significant). This result is consistent with Steele et al.'s (2003) findings showing improvements in children's theory of mind abilities that are associated with age.

In the forgiveness condition, comparison participants, as expected, used intent and apologies information for inferring willingness to forgive. This result is consistent with previous findings (e.g. Mullet et al., 2007). Persons with autism also used apologies information but they did not use intent information. For persons with autism, irrespective of their age, judging willingness to forgive seemed clearly more complex that judging of the appropriate level of blame to attribute. This new finding indicates that using intent information among persons with autism may be strongly task-dependent. In particular, it may be dependent on the type of relationship that is involved in the judgement task (direct or inverse). This finding is consistent with Cushman's (2008) findings showing that the impact of intent may strongly depend on the judgement task: namely, among adults without autism, intent was more strongly associated with blame judgement than it was with permissibility judgements.

Future studies, using a full range of moral situations (e.g. duty, envy, honesty, atonement, deserving, unkindness, unfairness, see Anderson, in press), and, in each case, a full range of responses types (e.g. blame, punishment, permissibility, see Cushman, 2008) should explore further the differences between persons with autism and comparison participants regarding the use of intent information, and the reasons for these differences.

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